



Experiment 10.2

```
#include <stdio.h>
#include <stdlib.h>
```

```
struct Node {
    int data;
    struct Node * next;
};
```

```
Void insertMiddle (struct Node * head, int position, int value) {
    struct Node * newNode = (struct Node *) malloc (size of (
        struct Node));
```

```
newNode -> data = value;
```

```
struct Node * temp = head;
int i;
```

```
for (i = 1; i < position && temp != NULL; i++) {
    temp = temp -> next;
```

```
}
```

```
if (temp == NULL) {
    printf ("Position out of range.\n");
    return;
```

```
}
```

Teacher's Signature : _____

```
newNode ->next = temp ->next;  
temp ->next = newNode;
```

```
printf("Inserted %d at position %d\n", value, position+1);  
}
```

```
void PrintList(struct Node* head) {  
    struct Node* temp = head;  
    printf("%d ->", temp ->data);  
    temp = temp ->next;  
}
```

```
printf("NULL\n");  
}
```

```
int main() {  
    struct Node* head = (struct Node*)malloc(sizeof(struct Node));  
    struct Node* second = (struct Node*)malloc(sizeof(struct Node));  
    struct Node* third = (struct Node*)malloc(sizeof(struct Node));
```

```
head ->data = 10;  
head ->next = second;
```

```
second ->data = 20;  
second ->next = third;
```

```
third ->data = 30;  
third ->next = NULL;
```



```
Printf ("Original Linked List :\n");  
PrintList(head);  
insertMiddle(head, 2, 25);
```

```
Printf ("In Linked List After Insertion :\n");  
PrintList(head);
```

```
return 0;  
}
```